

Chawin Mingsuwan

www.linkedin.com/in/chawinmingsuwan | <https://chawinmingsuwan.me/> | chawinmin@gmail.com | 818-599-9058

Education

Purdue University <i>Bachelor of Science in Video Game Development</i> <i>Minors in Advanced Global Technology, Earth and Planetary Science; Certification in Data Science</i> <i>Dean's List Fall 2023 - Fall 2025</i>	West Lafayette, IN May 2027 GPA: 3.96
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Professional Experience

NASA/Barrios - Purdue Datamine Corporate Partner <i>Lead User-Experience (UX) and Virtual Reality (VR) Unreal Engine Developer</i> - Led the integration of all interactive systems, rotation controls, dynamic color-coding, multi-camera logic, and UI panel interactions by unifying Blueprint and C++ behavior for mission planning and astronaut training. - Built a Meta Quest 2 VR simulation of the NASA Gateway station in Unreal Engine 5, enabling early astronaut training use-cases, including EVA planning and spatial orientation.	West Lafayette, IN January 2025 - December 2025
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Los Alamos National Laboratory <i>Extended Reality (XR) and High-Performance Computing (HPC) Intern</i> - Engineered immersive XR environments in Unreal Engine 5 to visualize high-performance computing simulations from ParaView to enable scientific research. - Implemented first-of-its-kind Apple Vision Pro hand gesture interactions using C++ and Blueprints, applying 3D joint vector math (distances and thresholds) to detect gestures and trigger real-time UI interactions.	Los Alamos, NM May 2025 - August 2025
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NASA L'SPACE Workforce Development Mission Concept Academy <i>Student Lead Computer Hardware Engineer and Scientist</i> Designed the Command and Data Handling subsystem and software flowchart for a conceptual lunar rover; performed TRL assessments and regolith research to ensure lunar environment readiness.	Remote September 2024 - December 2024
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Ilender Peru Internship <i>Software Engineering Intern</i> Automated multi-country data workflows (Peru, Bolivia, Mexico, Colombia, Venezuela) in Python and built an ARIMA forecasting model, improving data processing efficiency by 30% and prediction accuracy by 20%.	Lima, Peru May 2024 - July 2024
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Projects and Publications

NASA/IEEE Publication — Foundations of a Visualization Tool for NASA Gateway and Lunar Surface Operations First Author, peer-reviewed NASA/IEEE publication presented at the 2025 IEEE SMC-IT Conference, demonstrating XR visualization for Gateway mission planning.
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Johns Hopkins Applied Physics Laboratory (APL) — NASA JSC XR Mission Control Environment Implemented the VR interaction setup, early AI persona integration in Unity, and initiated a design interview with NASA astronaut Roy Bridges to ground the system in real operational workflows.

Tidal Disruption Event (TDE) Astrophysics Pipeline - Cleaned and processed ALeRCE/ZTF light-curve data in Python to identify TDE candidate signatures, including the characteristic $t^{-5/3}$ decay pattern. - Built and deployed a full-stack visualization pipeline in Python, implementing a FastAPI backend and frontend with JavaScript/React that generates Matplotlib light-curve plots.

Virtual Reality Study Room Prototype Led development of an interactive VR Study Room prototype in Unity, programming core C# mechanics to enhance user engagement and improve accessibility for alternative learning methods.

Leadership Experience

Purdue Space Day <i>Director of Activities and Treasurer</i> Coordinated 11+ orgs, 400+ volunteers, and 9 STEM activities for 600 students; first to integrate the NASA Gateway Visualization Tool (GVT) into the event.	West Lafayette, IN October 2024 - Present
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